## DIPOLE TYPE BY LENGTH
$l$ = antenna rod length (m)

$$\boxed{\lambda \text{ (wavelength)} = \frac{c}{f} = \frac{3\times10^8}{f} \text{ (m)}}$$

| $l/\lambda$ | Type | Use formula |
|---|---|---|
| $< 1/50$ | Hertzian (short) | HERTZIAN $S(R,\theta)$, $D=1.5$ |
| $= 1/4$ | Quarter-wave | ARB formula, $\pi l/\lambda = \pi/4$ |
| $= 1/2$ | Half-wave | half-wave shortcut, $D=1.64$, $R_{\text{rad}} = 73\,\Omega$ |
| $= 1$ | Full-wave | ARB, $D=2.41$, $R_{\text{rad}} = 199\,\Omega$ |
| other | Arbitrary | ARB formula |

## HERTZIAN DIPOLE — $S(R,\theta)$

**Far-field E,H phasors** *small $l \ll \lambda$*

$$\tilde{E}_\theta = \frac{jI_0kl\eta_0}{4\pi}\frac{e^{-jkR}}{R}\sin\theta, \quad \tilde{H}_\phi = \tilde{E}_\theta/\eta_0$$

**Power density formulas** *$l/\lambda < 1/50$; far field*

| | |
|---|---|
| Power density | $S(R,\theta) = S_{\max}\sin^2\theta$ (W/m$^2$) |
| Max pwr density | $S_{\max} = \dfrac{\eta_0\,k^2 I_0^2\,l^2}{32\pi^2R^2}$ (W/m$^2$) |

| | |
|---|---|
| $l$: rod length (m) | $I_0$: peak current (A) |
| $R$: obs. distance (m) | $\theta$: angle from dipole axis (rad) |
| $k = 2\pi/\lambda$ (rad/m) | $\eta_0 = 120\pi \approx 377\,\Omega$ |

**Total radiated power** $P_{\text{rad}} = \frac{\eta_0\pi}{3}(I_0 l/\lambda)^2$ (W)

## ARBITRARY-LENGTH DIPOLE — $S(\theta)$

**Power density at $R$** *any $l$*

$$S(\theta) = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

*At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow$ $S=0$ (no axial radiation). TI-36X alt: evaluate bracket at $\theta = 0.001$ rad $\to$ tiny $\to$ 0.*

**Normalized pattern** *dimensionless* $F(\theta) = S(\theta)/S_{\max}$

## COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

**Conversion** *multiply by $\pi/180$ or $180/\pi$*

$$\theta_{\text{rad}} = \theta_{\text{deg}}\cdot\frac{\pi}{180} \qquad \theta_{\text{deg}} = \theta_{\text{rad}}\cdot\frac{180}{\pi}$$

| $^\circ$ | rad | $\sin\theta$ | $\cos\theta$ | $\sin^2\theta$ | $\cos^2\theta$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 |
| 30 | $\pi/6$ | 1/2 | $\sqrt{3}/2$ | 1/4 | 3/4 |
| 45 | $\pi/4$ | $\sqrt{2}/2$ | $\sqrt{2}/2$ | 1/2 | 1/2 |
| 60 | $\pi/3$ | $\sqrt{3}/2$ | 1/2 | 3/4 | 1/4 |
| 90 | $\pi/2$ | 1 | 0 | 1 | 0 |
| 180 | $\pi$ | 0 | $-1$ | 0 | 1 |

*Antenna formulas usually take $\theta$ in rad. RAD mode on calc.*

## LINEAR ANTENNA ARRAY

**Array factor (uniform, broadside)** *$k = 2\pi/\lambda$, $\gamma = kd\cos\theta$*

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}; \quad F_a^{\max} = N^2 \text{ at } \theta = \pi/2$$

**Normalize** $F_a^{\text{norm}} = F_a/N^2$

**HPBW (broadside, uniform)** *use $Nd$, not $(N-1)d$*

$$\beta \approx 0.88\,\frac{\lambda}{Nd} \text{ (rad)} = 50.8^\circ\cdot\frac{\lambda}{Nd}$$

**Electronic steering (scan to $\theta_0$)** *replace $\gamma$ with $\gamma' = kd(\cos\theta - \cos\theta_0)$*

$$\delta = \frac{2\pi d}{\lambda}\cos\theta_0\ \left(\frac{\text{rad}}{\text{elt}}\right); \quad \theta_0 = \frac{\pi}{2}\text{:broadside, 0:end-fire}$$

**Frequency scan** *feed-line incr. $\Delta\ell = n_0\lambda_0$*

$$\cos\theta_0 \approx \frac{n_0\lambda_0}{d}\,\frac{\Delta f}{f_0}$$

*Grating lobes appear if $d > \lambda$. Avoid.*

## BEAM PARAMETERS — $\beta$, $\Omega_p$, $D$

**Normalize** *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

**Beamwidth $\beta$ at threshold $X$**

Set $F(\theta_X) = X$, solve for $\theta_X$; $\beta = 2\theta_X$ (symmetric)

| Threshold | $X = 10^{\text{dB}/10}$ | $\theta_X$ |
|---|---|---|
| HPBW ($-3$ dB) | 0.5 | $\sqrt{\frac{\ln 2}{a}}$ |
| $-6$ dB | 0.25 | $\sqrt{\frac{\ln 4}{a}}$ |
| $-10$ dB | 0.1 | $\sqrt{\frac{\ln 10}{a}}$ |
| any $F(\theta_X)$ | any $X$ | $F(\theta_X) = X$ |

**Pattern solid angle $\Omega_p$** *recognize pattern, look up*

$$\Omega_p = \int_0^{2\pi}\int_0^{\pi} F(\theta)\sin\theta\,d\theta\,d\phi \text{ (sr)}$$

*No $\phi$ dep.: $\Omega_p = 2\pi\int_0^\pi F(\theta)\sin\theta\,d\theta$.*

*$\theta$ is the integration variable — DON'T plug in a number for $\theta$ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0,\pi]$.*

| Pattern $F(\theta)$ | $\Omega_p$ (sr) |
|---|---|
| Isotropic ($F=1$) | $4\pi \approx 12.57$ |
| Hertzian $\sin^2\theta$ | $8\pi/3 \approx 8.38$ |
| Half-wave dipole | $\approx 7.66$ |
| Narrow Gauss $e^{-a\theta^2}$ | $\pi/a = \pi\theta_{1/2}^2/\ln 2$ |
| 2-axis HPBW (aperture) | $\beta_{xz}\cdot\beta_{yz}$ |
| Other (use integral) | numerical |

*TI-36X: `[2nd][d/dx]` for $\int_\square^\square$, RAD mode, multiply by $2\pi$.*

**Directivity $D$** *always $D = 4\pi/\Omega_p$; common shortcuts below*

$$D = \frac{4\pi}{\Omega_p} \text{ (unitless)} \qquad D_{\text{dB}} = 10\log_{10} D$$

*WORD TRAP: "direction" = $\theta_{\max}$ (rad); "directivity" = $D$ (unitless).*

**Gain (if asked)** *$\xi$ = radiation efficiency (unitless), $0 < \xi \le 1$*

$$G = \xi D \text{ (unitless)} \qquad G_{\text{dB}} = 10\log_{10} G \text{ (dB)}$$

*Lossless / ideal antenna $\Rightarrow$ $G = D$.*

## COMMON DIPOLE PARAMETERS

*Lossless ($\xi = 1$): $G = D$.*

| Type | $l$ | $D$ | $D_{\text{dB}}$ | $R_{\text{rad}}$ ($\Omega$) |
|---|---|---|---|---|
| Isotropic | N/A | 1 | 0 | — |
| Hertzian | $<\lambda/50$ | 1.5 | 1.76 | $80\pi^2(l/\lambda)^2$ |
| Half-wave | $\lambda/2$ | 1.64 | 2.15 | 73.2 |
| Monopole | $\lambda/4$ (gnd) | 1.64 | 2.15 | 36.6 |
| Full-wave | $\lambda$ | 2.41 | 3.82 | 199 |

*Half-wave: $P_{rad} = 36.6I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but $P_{rad}$, $R_{rad}$ halved.*

## ANTENNA AREAS ($A_e$, $A_p$)

**Effective area** *antenna's RX cross section; $D$ from BEAM PARAMS or table above*

$$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$$

**Physical cross section** *wire dipole, diameter $d$; $l$ see table above*

$$A_p = l\,d \text{ (m}^2\text{, any dipole)}$$

**Inverse:** $D$ from $A_e$

$$D = \frac{4\pi A_e}{\lambda^2} \text{ (unitless)}$$

*Thin wire: $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).*

## FRIIS TRANSMISSION FORMULA

*For each antenna's G: if given by NAME ("half-wave dipole" etc.) $\to$ COMMON DIPOLE PARAMS ($G=D$, lossless). Else (dB/linear value given) $\to$ use as given (convert dB $\to$ linear).*

**Pwr density at RX, then RX pwr** *$G_t, G_r$ linear gains; $\lambda = c/f$*

$$S_r = \frac{G_tP_t}{4\pi R^2} \text{ (W/m}^2\text{)}; \quad \boxed{P_r = G_tG_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t} \text{ (W)}$$

*Equivalent: $P_r = S_rA_{e,r}$.*

**Solve for $R$** *max range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_tG_rP_t}{P_r}} \text{ (m)}$$

**Equivalent form using $A_e$** *recv. area form*

$$P_r = \frac{G_tP_tA_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

*Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_tG_rP_t/L_{fs}$.*

*Off-axis (patterns not aligned): multiply by $F_t(\theta_t,\phi_t)F_r(\theta_r,\phi_r)$, both = 1 when aligned.*

**Recipe** 1. $\lambda = c/f$. 2. Convert any dB gain $\to$ linear: $G = 10^{G_{\text{dB}}/10}$. 3. Get $G_t, G_r$ by antenna type: dipole$\to$COMMON DIPOLE PARAMS; aperture$\to$APERTURE box; arbitrary $F\to$BEAM PARAMS. 4. Plug into Friis.

*Use linear G, not dB. Convert FIRST.*

## dB $\leftrightarrow$ LINEAR

*Works for any power-like quantity (G, D, $P_r/P_t$, SNR, ...):*

$$X_{\text{dB}} = 10\log_{10}X_{\text{lin}} \qquad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

| dB | linear | dB | linear |
|---|---|---|---|
| 0 | 1 | 10 | 10 |
| 3 | 2 | 13 | 20 |
| 6 | 4 | 20 | 100 |
| 7 | 5 | 30 | 1000 |
| $-3$ | 0.5 | $-10$ | 0.1 |

*Memorize: 3 dB $\Leftrightarrow \times2$, 10 dB $\Leftrightarrow \times10$. Add dB = multiply linear.*

*For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).*

## NOISE & SNR

**Noise power** *thermal noise into matched receiver*

$$P_N = k_B\,T_{\text{sys}}\,B \text{ (W)}$$

$k_B = 1.38\times10^{-23}$ J/K; $T_{\text{sys}}$ system noise temp (K); $B$ receiver bandwidth (Hz).

**Signal-to-noise ratio**

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N} \text{ (unitless)} \qquad \text{SNR}_{\text{dB}} = 10\log_{10}\frac{P_{\text{rec}}}{P_N}$$

*Typical comm. link needs SNR > 10 dB for usable signal.*

## APERTURE ANTENNAS (horn, dish)

*$A_p$ physical aperture area (m$^2$); $A_e$ effective area (m$^2$, $=\lambda^2D/4\pi$); $\beta_{xz}, \beta_{yz}$ HPBWs in two planes (rad); $\ell, d$ aperture side / diameter (m).*

**Directivity (any shape, uniform illum.)**

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \quad \Rightarrow \quad A_e \approx A_p$$

**Directivity from beamwidths** *any aperture*

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi/\beta^2$$

**HPBW and $D$ by aperture shape** *uniform illum.; $\beta$ in rad*

| Shape | $A_p$ | HPBW (rad) | $D$ (unitless) |
|---|---|---|---|
| Rect. $\ell_x\times\ell_y$ | $\ell_x\ell_y$ | $\beta_x \approx 0.88\lambda/\ell_x$<br>$\beta_y \approx 0.88\lambda/\ell_y$ | $D \approx 4\pi/(\beta_x\beta_y)$<br>$= 4\pi\ell_x\ell_y/\lambda^2$ |
| Square $\ell$ | $\ell^2$ | $\beta \approx 0.88\lambda/\ell$ | $D \approx 4\pi/\beta^2 = 4\pi\ell^2/\lambda^2$ |
| Circular (diam. $d$) | $\pi d^2/4$ | $\beta \approx 1.02\lambda/d$ | $D \approx 4\pi/\beta^2 = \pi^2d^2/\lambda^2$ |

**Scaling (any ratio)** *$D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$*

$$D_{\text{new}} = D_{\text{old}}\cdot\frac{A_{p,\text{new}}}{A_{p,\text{old}}}\cdot\left(\frac{f_{\text{new}}}{f_{\text{old}}}\right)^2$$

$$\beta_{\text{new}} = \beta_{\text{old}}\cdot\sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}}\cdot\frac{f_{\text{old}}}{f_{\text{new}}}$$

*If a parameter is unchanged, its ratio = 1. Examples: double area $\to$ $D\times2$, $\beta\times1/\sqrt{2}$; double freq $\to$ $D\times4$, $\beta\times1/2$.*

## DIPOLE

- $l$ — antenna length (m)
- $\lambda = c/f$ (m); $c = 3\times10^8$ m/s
- $I_0$ — peak current (A)
- $k = 2\pi/\lambda$ (rad/m)
- $R$ — obs. dist. (m); $\theta$ from axis
- $\eta_0 = 120\pi \approx 377\,\Omega$
- $S(R,\theta)$ — pwr density (W/m$^2$)

## BEAM

- $F(\theta) = S/S_{\max}$ (unitless)
- $a$ — coef. on $\theta^2$ in Gaussian
- $\beta$ — HPBW (rad or $^\circ$)
- $\Omega_p$ — pattern solid angle (sr)
- $D = 4\pi/\Omega_p$ — directivity
- $\xi$ radiation eff. (unitless); $G = \xi D$
- $D_{\text{dB}} = 10\log_{10}D$; $D = 10^{D_{\text{dB}}/10}$

## FRIIS / dB

- $P_t$ — power transmitted (W)
- $P_r$ — power received (W)
- $G_t$ — transmit gain (linear)
- $G_r$ — receive gain (linear)
- $S_r = G_tP_t/(4\pi R^2)$ (W/m$^2$)
- $P_r = G_tG_r(\lambda/4\pi R)^2P_t$
- $G = 10^{G_{\text{dB}}/10}$
- 3 dB $= \times2$, 10 dB $= \times10$
- $P_N = k_BT_{\text{sys}}B$ (W)

## APERTURE

- $A_p$ — physical area (m$^2$)
- $A_e = \lambda^2D/(4\pi)$ (m$^2$)
- $A_e \approx A_p$ for aperture
- $D \approx 4\pi A_p/\lambda^2$ (unitless)
- $\beta \approx 0.88\lambda/\ell$ (rect/sq.)
- $\beta \approx 1.02\lambda/d$ (circular)
- $2A_p \Rightarrow 2D$, $\beta/\sqrt{2}$

## ARRAY

- $N$ — number of elements
- $d$ — spacing (m)
- $\gamma = kd\cos\theta$
- $F_a = \sin^2(N\gamma/2)/\sin^2(\gamma/2)$
- $F_a^{\max} = N^2$ at $\theta = \pi/2$
- $F_a^{\text{norm}} = F_a/N^2$
- $\beta \approx 0.88\lambda/(Nd)$ (broadside)